

$$\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x = 338 \sin t. \quad [7]$$
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where the constants R and ϕ are to be determined. [3]

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7

[3]

[illegible]

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$$x \frac{dy}{dx} - y = x^2 \sqrt{x^2 - 9},$$

[9]

[illegible]